INCH-POUND

MIL-PRF-1/970E
27 August 1999
SUPERSEDING
MIL-E-1/970D
27 December 1976

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, GAS SWITCHING
TYPE 1B58A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: TR, bandpass, frequency range 2,664 to 2,964 MHz, incident power 750 kW.

ABSOLUTE RATINGS:

| Parameter:<br>Unit: | Incident power<br>kW | Ebb (open circuit)<br>V dc | Alt<br>ft |
|---|---|---|---|
| Maximum<br>Minimum | ---<br>10 | ---<br>-700 | 10,000<br>--- |

PHYSICAL CHARACTERISTICS: See figure 1.

TEST CONDITIONS:

| Parameter:<br>Unit: | F<br>MHz | tp1<br>µs | tp2<br>µs | prr<br>pps | Du<br>--- | Incident power<br>kW | Ii<br>µA dc | Ri<br>meg |
|---|---|---|---|---|---|---|---|---|
| Test condition 1:<br>Test condition 2:<br>Test condition 3: | F4 ±0.5%<br>F4 ±0.5%<br>F4 ±0.5% | 1.0 ± 0.15<br>1.0<br>--- | 0.5 ± 0.15<br>---<br>--- | 1,000<br>1,000<br>--- | ---<br>0.001<br>0.0008 | 200 ±10<br>50<br>750 ±10% | 200<br>200<br>200 | 1.6<br>1.6<br>1.6 |

| Test frequencies | |
|---|---|
| F | MHz tolerance ± 0.1% |
| 1 | 2664 |
| 2 | 2689 |
| 3 | 2714 |
| 4 | 2802 |
| 5 | 2914 |
| 6 | 2939 |
| 7 | 2964 |

GENERAL:

Qualification: Required.


DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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TABLE I. Testing and inspection.

| Requirement or test | Method | Test | Conditions | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | | | | | Min | Max | |
| Qualification inspection | | | | | | | |
| Degradation due to vibration | 4021 | --- | | --- | --- | --- | --- |
| High-level VSWR | 4774 | 2 | 2/ | --- | --- | 1.15 | --- |
| Conformance inspection, part 1 | | --- | 5/ | | | | |
| Ignitor ignition time | 4401 | --- | Ebb = -600 V dc | t | --- | 5.0 | sec |
| Ignitor voltage drop | 4406 | --- | Ii = 200 µA dc | Eid | 200 | 400 | V dc |
| Spike-leakage energy | 4452 | 1 | 2/ | Ws | --- | 0.3 | erg |
| Flat-leakage power | 4452 | 1 | 2/ | pf | --- | 40 | mw |
| Temperature cycling (nonoperating) | 1027 | --- | 1 cycle | --- | --- | --- | --- |
| Low-level VSWR | 4473 | --- | F1<br>F2 to F3<br>F3 to F5<br>F5 to F6<br>F7<br>1/ 3/ | ---<br>---<br>---<br>---<br>--- | ---<br>---<br>---<br>---<br>--- | 1.65<br>1.2<br>1.3<br>1.2<br>1.65 | ---<br>---<br>---<br>---<br>--- |
| Conformance inspection, part 2 | | | | | | | |
| Dielectric material strain | 4101 | --- | | --- | --- | --- | --- |
| Pressurizing | 4003 | --- | 50 $lb_f/in^2$ 4/ | --- | --- | --- | --- |
| Insertion loss (fixed tuned) | 4416 | --- | F4; Ii = 0 1/ | Li | --- | 0.3 | dB |
| Ignitor interaction (insertion loss) | 4421 | --- | Ii = 200 µA dc | Li | --- | 0.1 | dB |
| Recovery time | 4471 | 3 | Ii = 200 µA dc 2/ | t | --- | 15 | µs |
| Insertion | --- | --- | 6/ | --- | --- | --- | --- |

See footnotes at end of table.



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TABLE I. Testing and inspection - Continued.

| Requirement or test | Method | Test | Conditions | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | | | | | Min | Max | |
| Conformance inspection, part 3 | | | | | | | |
| Life test | --- | 3 | Group D; Ii = 150 to 200 µA dc 2/ | t | 500 | --- | hrs |
| Life test end points: | --- | | | | | | |
| Insertion loss (fixed tuned) | 4416 | --- | F4; Ii = 0 1/ | Li | --- | 1.0 | dB |
| Spike-leakage energy | 4452 | 1 | 2/ | Ws | --- | 0.3 | erg |
| Flat-leakage power | 4452 | 1 | 2/ | pf | --- | 40 | mW |
| Recovery time | 4471 | 3 | 2/ | t | --- | 30 | µs |
| Temperature cycling life-test end point | 1027 | --- | Group C; 10 cycles | --- | --- | --- | --- |

1/ This test shall be performed using the flanges specified in Drawing 268-JAN.

2/ This test shall be performed using the mount specified in Drawing 153-JAN.

3/ A swept frequency method of measurement may be used instead of measurement at fixed frequencies.

4/ The tube shall be mounted in accordance with Drawing 208-JAN, and the complete assembly cycled once from -55°C to +100°C. After the temperature has again reached room temperature, the pressure shall not have changed more than one-quarter of a pound/in$^2$ as indicated by a gauge permanently connected into the system.

5/ Unless otherwise specified, the acceptance level for all tests listed under conformance inspection, part 1, shall be 1.0, in accordance with the accept on zero (c = 0) sampling plans (Table III of MIL-PRF-1).

6/ The tube shall be capable of being inserted and removed from the mount specified in Drawing 153-JAN a minimum of 15 times with no deterioration in the tube's electrical characteristics.



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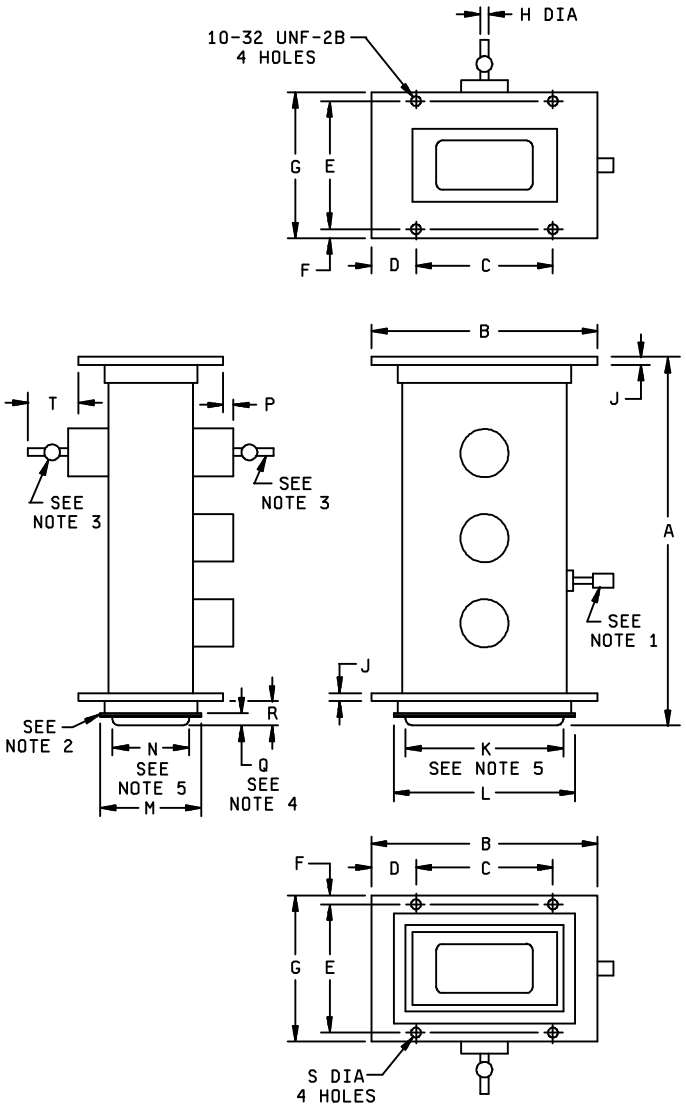


| Ltr | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| Conformance inspection, part 1 5/ | | | | |
| A | 6.580 | 6.640 | 167.13 | 168.66 |
| K | --- | 3.008 | --- | 76.40 |
| P | --- | .190 | --- | 4.83 |
| Q | .165 | .195 | 4.19 | 4.95 |
| R | .347 | .377 | 8.81 | 9.58 |
| T | --- | .810 | --- | 20.57 |
| Conformance inspection, part 2 | | | | |
| C | 2.490 | 2.510 | 63.25 | 63.75 |
| E | 2.240 | 2.260 | 56.90 | 57.40 |
| L | 3.203 | 3.233 | 81.36 | 82.12 |
| M | 1.703 | 1.733 | 43.26 | 44.02 |
| N | --- | 1.508 | --- | 38.30 |
| Reference dimensions | | | | |
| B | 4.12 | | 104.65 | |
| D | .81 | | 20.57 | |
| F | .19 | | 4.83 | |
| G | 2.62 | | 66.55 | |
| H | .25 | | 6.35 | |
| J | .12 | | 3.05 | |
| S | .22 | | 5.59 | |

NOTES:

1. Exhaust tube shall not extend beyond flange more than .25 inch (6.35 mm).
2. Gasket in accordance with Drawing 189-JAN. Gasket to be securely attached.
3. Ignitor electrode may be mounted in either of positions shown.
4. Dimension Q shall be measured prior to the attachment of the gasket to the tube.
5. The edges of the input window plate shall have either a radius of .020 inches (.51 mm) minimum or a 45° chamfer of .020 inch (.51 mm) minimum.
6. Mount either series or shunt mount may be used. If series mount is used, mount shall be in accordance with Drawing 153-JAN.
7. Unless otherwise specified, the acceptance level for all tests listed under conformance inspection, part 1, shall be 1.0, in accordance with the accept on zero (c = 0) sampling plans (Table III of MIL-PRF-1).

FIGURE 1. Outline drawing of electron tube type 1B58A.



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Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Navy - AS, CG, MC, OS, SH

Preparing activity:
DLA - CC

(Project 5960-3551-15)



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